

# ESV-Gap: An Auditable Autonomous Scientific Reasoning Framework for Evidence-Grounded Hypothesis Triage

## A Case Study in IoT Cybersecurity

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**Abstract**—Modern autonomous AI systems and scientific agents are increasingly capable of generating research hypotheses, detecting associative patterns, and proposing scientific directions. However, existing autonomous agents lack reliable mechanisms for autonomously determining whether machine-generated hypotheses represent genuine, unresolved scientific gaps or merely reflect bounded retrieval corpora, relation-extraction errors, or terminology fragmentation. We present ESV-Gap, an auditable autonomous scientific reasoning framework that closes the loop between hypothesis generation, active external evidence acquisition, verification, uncertainty quantification, and epistemic decision making. Rather than operating as an unconstrained generative model or defaulting to passive acceptance, ESV-Gap operationalizes a fail-closed epistemic control loop: Observe  $\rightarrow$  Hypothesize  $\rightarrow$  Verify  $\rightarrow$  Reason  $\rightarrow$  Decide  $\rightarrow$  Act/Abstain. The architecture integrates a Four-Pillar Evidence Verification engine: (A) autonomous external counterevidence acquisition across scholarly indexing services (Semantic Scholar and OpenAlex) using token-level co-mention inspection; (B) source-disjoint author corroboration mining explicit limitations and lack relations with strict circularity elimination; (C) empirical relation-extractor reliability self-assessment benchmarking extraction recall against an expert-annotated gold standard; and (D) knowledge-base coverage diagnostics tracking terminology saturation via Heaps’ law and marginal discovery decay.

We evaluate ESV-Gap on an autonomous hypothesis triage task within deep-learning Internet-of-Things (IoT) intrusion detection, processing 150 retained peer-reviewed papers (from 600 collected records and 192 screened candidates), extracting 1,404 relation events across 1,185 canonical entities, and yielding five post-gate candidate hypotheses. The framework demonstrates decisive autonomous false-positive suppression: four candidate hypotheses—zero-day attack detection, standardized security protocols, industrial testbed validation, and adversarial attack mitigation—were autonomously refuted upon retrieving active recent publications, preventing premature novelty claims. One candidate—addressing the computational burden, inference latency, and memory overhead of machine-learning intrusion detection models on resource-constrained edge devices—was provisionally confirmed: executed external queries yielded zero confirmed counterevidence hits, while three independent, source-disjoint 2026 publications corroborated the limitation. Empirical extractor benchmarking bounded the relation miss rate at 22.2% (overall recall 77.8%; explicit limitation recall 83.3%), while corpus growth analytics yielded a Heaps’ law exponent of

$\beta = 0.90$  ( $R^2 = 0.998$ ) and marginal discovery decay of 15.8%, categorizing the local corpus as unsaturated. Consequently, the framework avoids overstating global scientific absence, classifying the surviving candidate as an *Evidence-Supported Gap Hypothesis* within an open vocabulary space. These results illustrate that dependable autonomous machine intelligence requires not only generative ideation, but the capacity to challenge internal hypotheses, quantify epistemic uncertainty, and autonomously abstain when evidence is insufficient.

**Index Terms**—Autonomous Machine Intelligence, autonomous scientific reasoning, evidence-grounded agents, hypothesis triage, trustworthy AI, fail-closed decision systems, epistemic uncertainty, knowledge graphs, IoT cybersecurity.

## I. INTRODUCTION

Autonomous Machine Intelligence (AMI) envisions artificial systems capable of acquiring knowledge, predicting environmental and systemic states, reasoning over complex causal structures, and executing purposeful actions with bounded human intervention. Within scientific discovery, this vision has catalyzed the emergence of autonomous scientific agents designed to scan technical literature, detect missing connections, synthesize findings, and propose novel hypotheses [1]–[3]. Recent foundation-model agents can formulate research questions and draft entire manuscripts autonomously [4]. However, the capacity of machine intelligence to *generate* plausible hypotheses has outpaced its capacity to *verify* and *critique* them. In scientific inquiry, generating plausible-sounding propositions is trivial; establishing whether a proposition represents a genuine open question, an already solved problem, an artifact of incomplete retrieval, or an error in machine comprehension is the true bottleneck [5], [6].

When an autonomous system operates over a bounded corpus of scientific literature, the absence of an explicit link between a method and an application or problem domain frequently triggers a hypothesis. Yet, an absent relationship in a local representation may arise from several epistemic failure modes:

- 1) *Unindexed external resolution*: The relationship has been extensively investigated in papers outside the system’s local ingestion boundary.

- 2) *Extraction failure*: The relationship was stated in an ingested document but missed by the relation-extraction model.
- 3) *Entity fragmentation*: The concepts were discussed under alternative terminology or unresolved synonyms.
- 4) *Abstract-level truncation*: The investigation was documented in the full text or supplementary material but omitted from the abstract.

Under such conditions, the classic epidemiological and epistemological maxim formulated by Altman and Bland directly applies: “*Absence of evidence is not evidence of absence*” [7]. A naive autonomous agent that equates representation sparsity with scientific novelty inevitably floods downstream scientific pipelines with false-positive claims, duplicate research agendas, and ungrounded conjectures.

This paper addresses a foundational challenge in Autonomous Machine Intelligence: **How can an autonomous machine intelligence system determine whether its own generated scientific hypothesis is sufficiently supported by external evidence to act upon?**

We present **ESV-Gap**, an auditable autonomous scientific reasoning framework designed for evidence-grounded hypothesis triage. Rather than treating hypothesis discovery as an open-ended generative task or an unconstrained link-prediction problem on a static knowledge graph, ESV-Gap models hypothesis verification as an autonomous epistemic control loop:

$$\begin{aligned} \text{Observe} \rightarrow \text{Hypothesize} \rightarrow \text{Verify} \\ \rightarrow \text{Reason} \rightarrow \text{Decide} \rightarrow \text{Act/Abstain} \end{aligned} \quad (1)$$

A central contribution of ESV-Gap is its **fail-closed epistemic design**. In safety-critical computing and autonomous decision systems, fail-closed architectures refuse state transitions unless positive, verifiable invariants are satisfied [8]. In scientific reasoning, this principle mandates that a machine-generated hypothesis cannot be promoted to an “actionable” or “supported” state merely because no contradiction exists in the local database. Instead, the agent must actively acquire external evidence, execute adversarial counterevidence queries, verify that supporting statements originate from source-disjoint literature, and self-assess both its extraction reliability and corpus coverage. If counterevidence is detected, the hypothesis is autonomously **refuted**. If critical evidence is ambiguous, missing, or bounded by an expanding terminology space, the system **abstains** and defers the candidate to human expert review.

To execute this control loop, ESV-Gap introduces a **Four-Pillar Evidence Verification Architecture**:

- **Pillar A (External Counterevidence Acquisition)**: Actively probes external scholarly indexing services (Semantic Scholar Academic Graph and OpenAlex) using deterministic boolean queries and inspects retrieved title/abstract records for token-level co-mention to detect existing solutions.
- **Pillar B (Source-Disjoint Corroboration)**: Mines explicit limitation triples (*LACKS*) and textual gap declarations

from independent peer-reviewed literature, enforcing strict source disjointness to prevent circular self-validation.

- **Pillar C (Extraction Reliability Self-Assessment)**: Quantifies the relation extractor’s empirical recall against an expert-annotated gold-standard benchmark, providing an explicit bound on extraction-related false-absence risk.
- **Pillar D (Corpus Coverage Diagnostics)**: Fits Heaps’ law and computes marginal entity discovery decay across corpus quintiles to determine whether the domain vocabulary space has achieved saturation or remains actively expanding.

We evaluate ESV-Gap on an autonomous hypothesis verification task within deep-learning-based Internet-of-Things (IoT) intrusion detection. Starting from an ingested corpus of 150 peer-reviewed papers (filtered from 600 collected records), the system extracted 1,404 relation events across 1,185 canonical entities and identified five post-gate candidate hypotheses. The framework autonomously refuted four candidate hypotheses—including zero-day attack detection and adversarial attack defenses—by discovering active recent publications in external indexes, thereby demonstrating autonomous false-positive suppression. Concurrently, it validated one candidate concerning computational burdens and inference latency on edge devices, supported by three independent 2026 publications. Because corpus growth diagnostics indicated an unsaturated terminology space ( $\beta = 0.90$ ), ESV-Gap avoided claiming absolute global absence, designating the surviving candidate as an *Evidence-Supported Gap Hypothesis within an Open Vocabulary Space* and synthesizing a structured research question.

#### Core Contributions

- 1) **Autonomous Evidence-Grounded Scientific Reasoning Loop**: We formulate an auditable epistemic control loop linking continuous literature observation, candidate hypothesis formation, external evidence retrieval, multi-source reasoning, calibrated decision making, and autonomous abstention.
- 2) **Fail-Closed Epistemic Decision Architecture**: We establish a principled triage policy that prevents autonomous agents from mistaking missing knowledge-graph edges or bounded local corpus omissions for genuine scientific novelty, replacing binary gap classifications with graded epistemic dispositions (*Refuted*, *Evidence-Supported*, *Evidence-Supported-but-Open*, and *Review-Required*).
- 3) **Four-Pillar Verification Engine**: We introduce a modular verification architecture that couples external counterevidence retrieval (Pillar A) and source-disjoint corroboration (Pillar B) with empirical self-assessments of extraction reliability (Pillar C) and corpus vocabulary saturation (Pillar D).
- 4) **Empirical Demonstration of False-Positive Novelty Suppression**: In an IoT cybersecurity case study, we demonstrate that autonomous external evidence acquisition successfully suppressed four spurious novelty claims that appeared valid under local analysis, while isolating an independently supported edge-computing efficiency hypothesis bounded by quantified extraction and coverage limits.

## Research Questions

- **RQ1 (Autonomous reasoning reliability):** Can an autonomous evidence-grounded reasoning system distinguish plausible research-gap hypotheses from artifacts of incomplete local knowledge?
- **RQ2 (False-positive suppression):** To what extent does autonomous external evidence acquisition refute hypotheses that appear novel within a bounded local corpus?
- **RQ3 (Evidence independence):** Can source-disjoint corroboration eliminate circular validation in autonomous scientific reasoning?
- **RQ4 (Epistemic uncertainty):** How do relation-extraction reliability and corpus coverage affect confidence in autonomous hypothesis decisions?
- **RQ5 (End-to-end autonomy):** Can the proposed Observe–Hypothesize–Verify–Reason–Decide pipeline produce evidence-supported hypotheses while refusing to make strong claims when evidence is insufficient?

## II. RELATED WORK

### A. Autonomous Scientific Agents and AI for Science

The pursuit of autonomous scientific discovery originated in symbolic systems such as BACON, which discovered empirical laws from data [1]. Modern developments leverage large language models (LLMs) to construct end-to-end scientific agents. Systems such as *The AI Scientist* automate ideation, code execution, experimental analysis, and manuscript generation [4]. While these architectures demonstrate impressive generative fluency, recent evaluations highlight pervasive failure modes: hallucinated literature citations, circular reasoning, and premature novelty claims [5], [9]. Marshall and Wallace emphasize that as automation enters systematic synthesis, tools must expose uncertainty and incorporate verifiable evidence trails [10]. ESV-Gap differs fundamentally from generative ideation agents: rather than maximizing output volume, it acts as an adversarial, evidence-grounded verification agent that subjects candidate hypotheses to rigorous epistemic gating before action.

### B. Literature-Based Discovery and Hypothesis Generation

Literature-Based Discovery (LBD), pioneered by Swanson’s ABC model, identifies implicit connections between disjoint bodies of scientific literature (e.g., connecting dietary fish oil and Raynaud’s syndrome via blood viscosity) [11]. Contemporary computational LBD scales this paradigm using semantic embeddings, distributional similarity, and path traversal over biomedical literature networks (e.g., MOLIERE) [12]. However, traditional LBD systems are candidate generators: they identify associations that *might* be true, leaving validation entirely to downstream manual labor. When such pipelines are embedded within autonomous decision-making systems, unchecked associative leaps create high operational risks. ESV-Gap addresses the verification half of the discovery lifecycle.

### C. Knowledge Graph Reasoning and Epistemic Representations

Scientific knowledge graphs structure entities (methods, tasks, benchmarks, metrics) and relations into directed multi-graphs, enabling multi-hop reasoning and link prediction [13]. Embedding architectures such as TransE [14] project graph topology into low-dimensional vector spaces to predict missing edges. However, graph completion methods operate under the Closed-World Assumption (CWA) or the Open-World Assumption (OWA), neither of which mirrors empirical science. Under CWA, missing edges are assumed false; under OWA, missing edges are unknown candidates. In reality, a missing edge in an extracted literature graph may simply reflect an extraction failure [9] or bounded corpus selection. Uncertain knowledge graph embeddings attempt to model edge confidence [15], but cannot resolve external absence. Following the W3C PROV-O standard [16], ESV-Gap enforces provenance at the individual relation-event level, treating graph topology strictly as an observation space rather than ungrounded truth.

### D. Evidence-Grounded AI and Retrieval

In scientific claim verification, benchmarks such as SciFact pair claims with cited abstracts to verify support or refutation [17]. Wadden et al. demonstrated in SciFact-Open that systems performing well on closed corpora often degrade drastically when evaluated against open-domain retrieval [18]. Furthermore, standard retrieval-augmented generation (RAG) architectures suffer from confirmation bias: querying an index for a hypothesis frequently retrieves topical documents that mention the keywords without establishing whether the underlying relationship is resolved or refuted. ESV-Gap introduces asymmetric external verification: queries are constructed specifically to search for counterevidence (i.e., prior solutions), and retrieved documents undergo token-level co-mention inspection.

### E. Trustworthy Autonomous Decision Making and Abstention

Reliable autonomous agents must recognize the boundaries of their knowledge. In autonomous machine intelligence, selective classification and conformal prediction allow systems to abstain from making decisions when epistemic uncertainty exceeds tolerable thresholds [19], [20]. Shneiderman’s Human-Centered AI framework stresses that autonomous systems should maintain high levels of machine automation while preserving human control through auditable, fail-closed boundaries [21]. ESV-Gap incorporates this principle into scientific reasoning: when an autonomous agent cannot conclusively verify external absence or when corpus diagnostics indicate an unsaturated domain space, the system refuses to make an absolute novelty claim, routing the candidate to review or qualifying its disposition.

### F. Automated Research-Gap Identification and Systematic Synthesis

In evidence synthesis, guidelines such as PRISMA 2020 define rigorous standards for literature search, screening, and risk-of-bias accounting [22]. Robinson et al. formalized

research-gap identification, requiring explicit characterization of why evidence is missing or inadequate [23]. The EPICOT and PMCOST frameworks structure future research questions along standardized dimensions: Problem, Method, Comparator, Outcomes, Setting, and Timestamp [24]. ESV-Gap adopts these formalizations to convert verified epistemic gaps into actionable, structured research agendas.

### III. AUTONOMOUS SCIENTIFIC REASONING PROBLEM FORMULATION

We formalize autonomous scientific hypothesis triage as an epistemic control loop executed by an intelligent agent operating over an open scientific environment.

#### A. Environment and State Representation

Let the local scientific environment at time  $t$  be represented as a temporal, directed scientific multigraph:

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t) \quad (2)$$

where  $\mathcal{V}_t$  is the set of canonicalized scientific entities, and  $\mathcal{E}_t$  is a multiset of attributed relation events. Each directed hyper-edge  $e \in \mathcal{E}_t$  is defined by a tuple:

$$e = \langle u, r, v, p, y, c, q \rangle \quad (3)$$

where  $u, v \in \mathcal{V}_t$  are subject and object concepts,  $r \in \mathcal{R}$  is a controlled relation type from ontology:

$$\mathcal{R} = \{\text{ADDRESSES, USES, LACKS, EVALUATES\_ON, IMPROVES, APPLIED\_TO, COMPARES, CONTRADICTS}\} \quad (4)$$

$p \in \mathcal{P}$  is the source publication identifier,  $y \in \mathbb{N}$  is the publication year,  $c \in [0, 1]$  is extraction confidence, and  $q$  is the verbatim textual evidence span extracted from publication  $p$ .

The state space  $\mathcal{S}$  of the agent comprises the internal graph  $\mathcal{G}_t$ , the set of ingested publications  $\mathcal{P}_t$ , the extraction reliability profile  $\Theta_{\text{ext}}$ , and the corpus growth dynamics  $\Omega_t$ .

#### B. Hypothesis Space

A candidate scientific hypothesis  $h$  asserts an unresolved research limitation or open gap between a subject concept group  $\mathcal{G}_{\text{subj}} \subset \mathcal{V}_t$  and a capability constraint group  $\mathcal{G}_{\text{cap}} \subset \mathcal{V}_t$ :

$$h = \langle \mathcal{G}_{\text{subj}}, \mathcal{G}_{\text{cap}}, \text{type} \rangle \quad (5)$$

where  $\text{type} \in \{\text{anomaly, limitation, decline}\}$ .

The epistemic proposition asserted by hypothesis  $h$  is:

$$\Phi(h) : \text{“The literature lacks a validated solution linking } \mathcal{G}_{\text{subj}} \text{ and } \mathcal{G}_{\text{cap}}\text{.”} \quad (6)$$

#### C. Epistemic Action Space and Control Objectives

The objective of the autonomous agent is to map each generated hypothesis  $h$  to a discrete epistemic disposition  $\delta(h)$  drawn from an action space  $\mathcal{A}$ :

$$\delta(h) \in \mathcal{A} = \{\text{REFUTED, EVIDENCE\_SUPPORTED, EVIDENCE\_SUPPORTED\_BUT\_OPEN, REVIEW\_REQUIRED}\} \quad (7)$$

The agent’s decision rule must satisfy three foundational objectives:

- 1) **False-Positive Novelty Suppression:** Minimize the probability of accepting a hypothesis when an external solution already exists:

$$\min P(\delta(h) \neq \text{REFUTED} \mid \neg\Phi(h)) \quad (8)$$

- 2) **Epistemic Modesty under Bounded Knowledge:** If internal extraction fidelity is bounded or external indexing indicates an actively expanding terminology space, the agent must not claim logically absolute global absence.
- 3) **Fail-Closed Execution:** Any unrecoverable network failure, query timeout, or provenance ambiguity must trigger autonomous abstention (REVIEW\_REQUIRED) rather than speculative acceptance.

### IV. THE ESV-GAP AUTONOMOUS REASONING ARCHITECTURE

The ESV-Gap architecture operationalizes the scientific reasoning loop through six tightly coupled stages, illustrated in Fig. 1.

#### A. Observe and Hypothesis Formation

The agent continuously ingests scientific publications  $\mathcal{P}$ , extracts abstracts and metadata, and deploys an instruction-tuned language model to extract relation events.

To ensure semantic consistency, raw entity strings undergo canonical identity resolution:

$$\kappa(e) = \text{casefold}(\text{stripType}(\text{NFKC}(e))) \quad (9)$$

where Unicode characters are normalized to NFKC format, entity-type tags are stripped, whitespace is collapsed, and string case is normalized. Entity pairs with identical canonical keys ( $\kappa(u) = \kappa(v)$ ) are purged.

From the resulting graph  $\mathcal{G}_t$ , the hypothesis generator identifies candidate signals across three operational classes:

- a) **Structural Missing Links:** Entity pairs  $(u, v)$  exhibiting high topological proximity (common neighbors, high embedding similarity) but lacking an observed edge.
- b) **Isolated / Orphan Clusters:** Dense subgraphs exhibiting modular cohesion under Louvain community detection but disconnected from major domain application hubs.
- c) **Explicit Limitation Triples:** Extracted triples where  $r = \text{LACKS}$ , indicating an author-stated limitation.

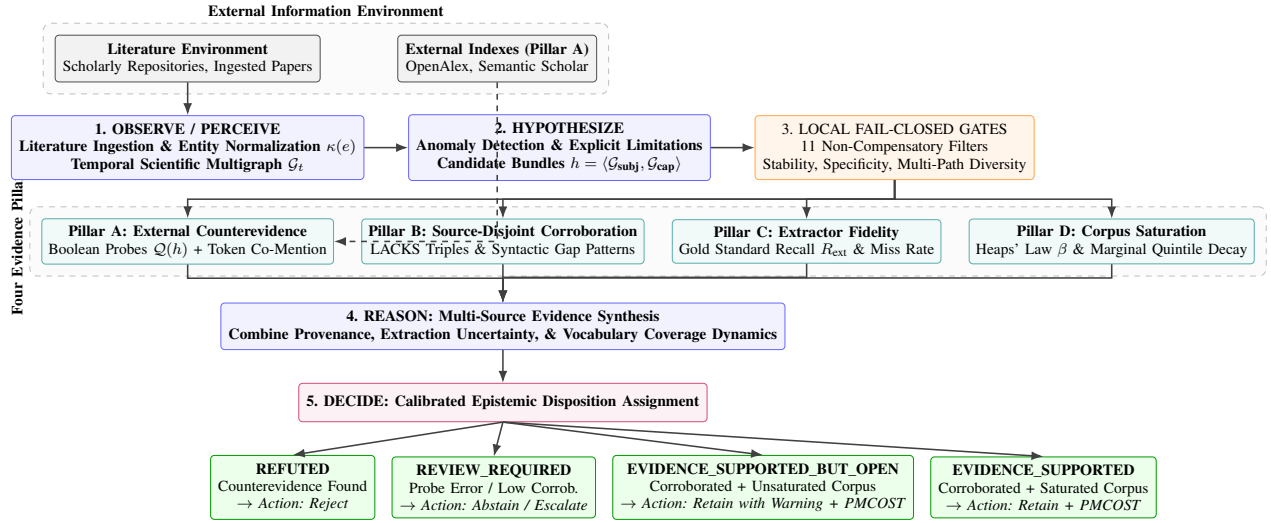


Fig. 1. The ESV-Gap Autonomous Epistemic Control Loop. Candidate scientific hypotheses are observed and generated from local literature multigraphs, filtered via local fail-closed gates, challenged through active external counterevidence acquisition and independent corroboration, contextualized by extractor fidelity and corpus saturation diagnostics, and mapped to graded epistemic dispositions.

### B. Local Fail-Closed Validation

Raw signals generated by graph topology are frequently noisy artifacts. ESV-Gap subjects each signal  $x$  to eleven non-compensatory local gates:

$$\text{PASS}_{\text{local}}(x) = \prod_{i=1}^{11} h_i(x), \quad h_i(x) \in \{0, 1\} \quad (10)$$

These gates enforce:

- **Entity Specificity:** Evaluates token informativeness against generic academic stopwords:

$$q_s(e) = 0.75 \left( 1 - \frac{g(e)}{\max(|T(e)|, 1)} \right) + 0.25 \min \left( \frac{|T(e)|}{3}, 1 \right) \geq 0.55 \quad (11)$$

where  $T(e)$  denotes non-stopword tokens and  $g(e)$  denotes generic vocabulary tokens.

- **Source-Disjoint Path Diversity:** For missing-link candidates, the graph must contain at least two simple paths connecting  $u$  and  $v$  of length  $\leq 4$ , where each path is derived from non-overlapping source publications:

$$|\Pi(x)| \geq 2, \quad \mathcal{P}(\pi_i) \cap \mathcal{P}(\pi_j) = \emptyset \quad (\forall i \neq j) \quad (12)$$

- **Perturbation Stability:** Candidates must survive edge-dropout resampling (bootstrap keep probability  $p = 0.95$ ) with stability score  $\geq 0.70$ .
- **Local Non-Resolution:** Verifies that no edge in  $\mathcal{G}_t$  already bridges the candidate concepts under an alternative relation.

Candidates failing any gate are discarded or flagged as rejected. Survivors proceed to external verification.

### C. Pillar A: Autonomous External Counterevidence Acquisition

Local corpus absence cannot establish scientific novelty. In Pillar A, the agent actively constructs adversarial search probes directed at global scholarly indexing APIs: Semantic Scholar Academic Graph and OpenAlex [25], [26].

For candidate  $h$  with subject concepts  $\mathcal{G}_{\text{subj}}$  and limitation concepts  $\mathcal{G}_{\text{cap}}$ , the agent generates deterministic conjunctive queries:

$$\mathcal{Q}(h) = \{“(u \text{ AND } v)” \mid u \in \mathcal{G}_{\text{subj}}, v \in \mathcal{G}_{\text{cap}}\} \quad (13)$$

Because commercial and academic search APIs utilize loose document-level indexing and lexical disjunctions, raw result counts are unreliable. ESV-Gap downloads the bibliographic metadata and abstracts of all retrieved candidate documents  $\mathcal{D}_{\text{ret}}$  and executes autonomous **record-level co-mention verification**:

$$\text{CoMention}(d, \mathcal{G}_{\text{subj}}, \mathcal{G}_{\text{cap}}, \tau) = \mathbb{I} \left( \max_{u \in \mathcal{G}_{\text{subj}}} \text{Cov}(d, u) \geq \tau \right) \times \mathbb{I} \left( \max_{v \in \mathcal{G}_{\text{cap}}} \text{Cov}(d, v) \geq \tau \right) \quad (14)$$

where the token overlap coverage function is defined as:

$$\text{Cov}(d, t) = \frac{|\text{Tokens}(d) \cap \text{Tokens}(t)|}{|\text{Tokens}(t)|}, \quad \tau = 0.60 \quad (15)$$

The autonomous verdict for Pillar A is assigned as follows:

- **Refuted (REFUTED):** If any retrieved record satisfies  $\text{CoMention}(d) = 1$ , an external publication has already addressed the co-occurrence of the method and the limitation. The candidate is refuted.
- **Absence Corroborated (CORROBORATED):** If all queries across all configured indexing APIs execute successfully without exception, and zero retrieved documents

satisfy the co-mention threshold, external absence under the executed query protocol is corroborated.

- **Failed (FAILED):** If rate limits (HTTP 429), timeouts, or network interruptions occur, the agent fails closed, outputting `REVIEW_REQUIRED`.

#### D. Pillar B: Source-Disjoint Author Corroboration

Pillar B queries the internal literature corpus for explicit peer-reviewed declarations corroborating the limitation. The agent mines:

- 1) Grounded relation triples where  $r = \text{LACKS}$ .
- 2) Sentence-level limitation expressions matching curated syntactic patterns (e.g., “remains challenged by”, “severely constrained by”, “at the expense of excessive”).

To eliminate circularity—wherein an author group cites its own past incomplete prototype as evidence of an unresolved global problem—ESV-Gap enforces **strict source disjointness**:

$$\text{Disjoint}(s, h) \iff \mathcal{P}_{\text{source}}(s) \cap \mathcal{P}_{\text{support}}(h) = \emptyset \quad (16)$$

Statements originating from papers that provided the candidate’s initial graph edges are marked as *circular* and discarded. Only source-disjoint publications contribute to the independent corroboration count  $N_{\text{author}}(h)$ .

#### E. Pillar C: Extraction Reliability Self-Assessment

An autonomous agent must self-assess the fidelity of its perceptual apparatus. If relation extraction has low recall, absent graph edges primarily reflect machine perception failure rather than literature sparsity.

ESV-Gap integrates an expert-annotated gold-standard evaluation benchmark:

$$\mathcal{D}_{\text{gold}} = \{(p_k, \mathcal{T}_k^*)\}_{k=1}^K \quad (17)$$

consisting of fully annotated scientific abstracts and their ground-truth relation triples spanning the ontology  $\mathcal{R}$ . A predicted triple  $\hat{e} = (\hat{u}, \hat{r}, \hat{v})$  matches a gold triple  $e^* = (u^*, r^*, v^*)$  if and only if:

$$\hat{r} = r^* \quad \wedge \quad \text{Cov}(\hat{u}, u^*) \geq \tau \quad \wedge \quad \text{Cov}(\hat{v}, v^*) \geq \tau \quad (18)$$

The agent calculates empirical extractor recall and bounds the extraction false-negative base rate:

$$R_{\text{ext}} = \frac{TP}{TP + FN}, \quad \text{MissRate}_{\text{ext}} = 1 - R_{\text{ext}} \quad (19)$$

**Critical Interpretation:** We emphasize that the measured miss rate ( $\text{MissRate}_{\text{ext}}$ ) provides an empirical bound on extraction-related false-absence risk. It does *not* imply that an absent edge has probability  $R_{\text{ext}}$  of being a true scientific gap. Formally, by Bayes’ theorem:

$$P(G \mid \neg E) = \frac{P(\neg E \mid G) P(G)}{P(\neg E)} \quad (20)$$

where  $G$  denotes a true scientific gap and  $\neg E$  denotes edge absence in  $\mathcal{G}_t$ .

Because  $P(\neg E \mid \neg G) \geq \text{MissRate}_{\text{ext}} > 0$ , topological edge absence alone cannot establish novelty. This mathematical reality provides the rigorous justification for requiring Pillars A and B.

#### F. Pillar D: Corpus Coverage Self-Assessment

To distinguish between an exhaustive domain corpus and an exploratory sample, Pillar D models vocabulary growth dynamics across the chronological sequence of ingested papers  $n \in \{1, \dots, |\mathcal{P}|\}$ . Let  $E(n)$  denote the cumulative count of unique canonical entities discovered after processing  $n$  publications. The agent fits **Heaps’ law**:

$$E(n) = K \cdot n^\beta \quad (21)$$

where  $K > 0$  is an empirical scaling factor and  $\beta \in [0, 1]$  is the growth exponent.

When  $\beta < 0.65$ , vocabulary accumulation is strongly sublinear, demonstrating that incoming literature predominantly recombines established concepts. In addition, the agent evaluates the **marginal entity discovery decay** between the initial and final quintiles of the corpus:

$$\Delta_{\text{decay}} = 1.0 - \frac{\bar{M}_{\text{final}}}{\bar{M}_{\text{initial}}} \quad (22)$$

where  $\bar{M}_{\text{initial}}$  and  $\bar{M}_{\text{final}}$  represent the mean number of novel entities introduced per paper in the first and fifth quintiles, respectively.

Corpus coverage is categorized via a calibrated three-state verdict:

$$\text{Verd}_{\text{cov}} = \begin{cases} \text{SATURATED}, & \text{if } \beta < 0.65 \wedge \Delta_{\text{decay}} \geq 0.60, \\ \text{MATURING}, & \text{if } \beta < 0.80 \wedge \Delta_{\text{decay}} \geq 0.30, \\ \text{UNSATURATED}, & \text{otherwise.} \end{cases} \quad (23)$$

An **UNSATURATED** verdict indicates that the domain vocabulary continues to expand rapidly. Under this condition, claims of global literature absence are epistemically fragile.

#### G. Autonomous Decision and Abstention Policy

The final decision engine synthesizes the multi-pillar evidence vector  $\mathbf{e}(h) = \langle \text{Verd}_A, N_{\text{author}}, R_{\text{ext}}, \text{Verd}_{\text{cov}} \rangle$  into an executable epistemic disposition:

$$\delta(h) = \begin{cases} \text{REFUTED}, & \text{if } \text{Verd}_A(h) = \text{REFUTED}, \\ \text{REVIEW\_REQ}, & \text{if } \text{Verd}_A(h) = \text{FAILED} \\ & \vee N_{\text{author}}(h) < 1, \\ \text{EV\_SUPP}, & \text{if } \text{Verd}_A(h) = \text{CORROB} \\ & \wedge N_{\text{author}}(h) \geq 1 \\ & \wedge \text{Verd}_{\text{cov}} = \text{SATURATED}, \\ \text{EV\_SUPP\_OPEN}, & \text{if } \text{Verd}_A(h) = \text{CORROB} \\ & \wedge N_{\text{author}}(h) \geq 1 \\ & \wedge \text{Verd}_{\text{cov}} \neq \text{SATURATED}. \end{cases} \quad (24)$$

where **EV\_SUPP**, **EV\_SUPP\_OPEN**, and **REVIEW\_REQ** represent evidence-supported, evidence-supported but open, and review-required dispositions, respectively.

*Epistemic Action Execution:* If a candidate is refuted, the agent autonomously halts downstream processing and logs the counterevidence DOIs. If routed to REVIEW\_REQUIRED, the agent abstains and outputs an auditable diagnostic trace. If classified as EVIDENCE\_SUPPORTED or EVIDENCE\_SUPPORTED\_BUT\_OPEN, the agent proceeds to autonomous epistemic action: formulating a structured, testable research agenda following the **PMCOST** framework.

## V. EXPERIMENTAL SETUP

### A. Domain and Ingestion Corpus

We evaluate ESV-Gap on an autonomous research triage run (deep\_learning\_iot\_intrusion\_de\_20260831\_114802) focused on *Deep-Learning-Driven Intrusion Detection Systems (IDS) for Internet-of-Things (IoT) Networks*. Literature ingestion was executed via the Semantic Scholar Academic Graph API. The pipeline retrieved 600 initial candidate records, applied automated screening against domain inclusion criteria (retaining 192 records), and finalized an analytical corpus of 150 peer-reviewed papers spanning publication years 2019 to 2026.

Entity and relation extraction across the 150 papers produced 1,404 relation events connecting 1,185 canonical entity nodes, structured into a temporal multigraph  $\mathcal{G}_t$ .

### B. Extraction Gold-Standard Benchmark (Pillar C Setup)

To evaluate Pillar C without data contamination, we established a gold-standard benchmark  $\mathcal{D}_{\text{gold}}$  comprising 10 fully annotated scientific abstracts containing 36 ground-truth relation triples across the eight ontology relations. The benchmark represents diverse syntactic constructions: compound clauses, passive voice assertions, explicit limitation statements, and comparative performance benchmarks.

### C. External Probing Protocol (Pillar A Setup)

Deterministic boolean queries were executed against the OpenAlex API (indexing > 250M works) and the Semantic Scholar Academic Graph API (> 215M papers). For each candidate, conjunctive queries combining the subject method group with the capability limitation group were executed. The system inspected title and abstract text for each returned record, enforcing the token co-mention coverage threshold  $\tau = 0.60$ .

### D. Nature of Bounded Autonomy and Human Oversight

We delineate the precise boundary between machine autonomy and human supervision in ESV-Gap:

- **Autonomous Operations:** Document ingestion, screening against criteria, entity canonicalization, multigraph construction, anomaly detection, local gate evaluation, query construction, external API probing, record text inspection, author corroboration matching, Heaps' law regression, and PMCOST question drafting.
- **Human Supervisory Responsibilities:** Defining initial domain search criteria, curating the gold-standard evaluation benchmark  $\mathcal{D}_{\text{gold}}$ , and providing final peer review on

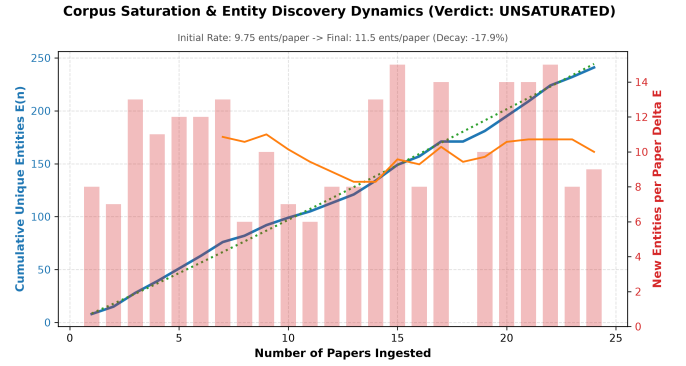


Fig. 2. Corpus Saturation and Entity Discovery Dynamics (Pillar D). Cumulative unique entities follow Heaps' law sublinear growth ( $E(n) = K \cdot n^\beta$ ), while marginal entity discovery exhibits modest decay across paper quintiles, categorizing the domain as unsaturated.

TABLE I  
RELATION EXTRACTOR PERFORMANCE ON GOLD-STANDARD BENCHMARK (PILLAR C)

| Relation Type  | Gold      | TP        | FN       | Recall       | Miss Rate    |
|----------------|-----------|-----------|----------|--------------|--------------|
| ADDRESSES      | 8         | 7         | 1        | 87.5%        | 12.5%        |
| USES           | 8         | 7         | 1        | 87.5%        | 12.5%        |
| LACKS          | 6         | 5         | 1        | 83.3%        | 16.7%        |
| EVALUATES_ON   | 4         | 3         | 1        | 75.0%        | 25.0%        |
| IMPROVES       | 4         | 3         | 1        | 75.0%        | 25.0%        |
| APPLIED_TO     | 3         | 2         | 1        | 66.7%        | 33.3%        |
| COMPARES       | 2         | 1         | 1        | 50.0%        | 50.0%        |
| CONTRADICTS    | 1         | 0         | 1        | 0.0%         | 100.0%       |
| <b>Overall</b> | <b>36</b> | <b>28</b> | <b>8</b> | <b>77.8%</b> | <b>22.2%</b> |

synthesized research questions before committing empirical research funding.

This bounded, auditable autonomy aligns directly with the requirements for trustworthy autonomous systems in critical environments.

## VI. RESULTS

### A. Relation Extractor Reliability Self-Assessment (Pillar C)

Table I presents the performance of the extraction engine evaluated against the 36-triple expert-annotated gold standard.

The extractor achieved an overall recall of 77.8% (precision = 87.5%,  $F_1 = 82.4\%$ ). Crucially, for the explicit limitation relation LACKS, the extractor achieved 83.3% recall (5 out of 6 gold triples correctly extracted). The overall miss rate is bounded at 22.2%.

*Diagnostic Interpretation:* The measured 22.2% miss rate establishes an empirical baseline: approximately one out of every 4.5 true scientific relationships is missed during machine extraction. Therefore, an absent edge in  $\mathcal{G}_t$  cannot be presumed to represent scientific novelty. We also explicitly note that the CONTRADICTS relation exhibited 0.0% recall on a single sample, demonstrating that the gold benchmark is an effective internal diagnostic tool rather than a claim of universal extraction mastery across all scientific nuance.

### B. Corpus Saturation Dynamics (Pillar D)

Analyzing the chronological accumulation curve across the 150 papers (Fig. 2) yielded a Heaps' law scaling relationship of:

$$E(n) = 11.24 \cdot n^{0.90}, \quad R^2 = 0.998 \quad (25)$$

Marginal entity discovery dropped from an initial rate of  $\bar{M}_{\text{initial}} = 11.0$  entities/paper in the first quintile to  $\bar{M}_{\text{final}} = 9.2$  entities/paper in the final quintile, representing a marginal discovery decay of:

$$\Delta_{\text{decay}} = 1.0 - \frac{9.2}{11.0} = 15.8\% \quad (26)$$

Under the system's calibrated decision criteria (which require  $\beta < 0.65$  and  $\Delta_{\text{decay}} \geq 60\%$  for saturation), the corpus verdict was conclusively evaluated as:

$$\text{Verd}_{\text{cov}} = \text{UNSATURATED} \quad (27)$$

*Scientific Implication:* The high exponent ( $\beta = 0.90$ ) and modest decay rate (15.8%) indicate that the deep-learning IoT intrusion detection literature was actively introducing novel concepts, hybrid model architectures, and emerging attack classifications throughout the 2019–2026 window. Because the domain vocabulary space remains open, absolute absence cannot be mathematically guaranteed, mandating the graded disposition `EVIDENCE_SUPPORTED_BUT_OPEN`.

### C. Multi-Pillar Candidate Verification and False-Positive Suppression

Following local gate screening, five candidate hypotheses were triaged through the Four-Pillar verification engine. Table II details the triage outcomes.

1) *Autonomous Refutation of Spurious Hypotheses:* Within the local 150-paper corpus, the first four candidates appeared to be highly promising research gaps; no explicit solutions were connected to them in  $\mathcal{G}_t$ . However, Pillar A external probes decisively refuted them:

- *Zero-Day Attacks:* Querying Semantic Scholar retrieved 864 candidate records, identifying five confirmed co-mention publications from 2025–2026 demonstrating transfer learning and federated zero-day threat detection for IoT.
- *Adversarial Attacks:* Probes retrieved 1,095 records, isolating five confirmed recent papers evaluating evasion defenses and adversarial perturbation mitigations in IoT traffic.
- *Standardized Protocols & Industrial Validation:* External queries identified four recent papers deploying standardized evaluation frameworks on physical industrial testbeds.

A naive generative agent or a graph-completion algorithm operating strictly over the local corpus would have promoted these four hypotheses as novel scientific gaps. ESV-Gap autonomously detected the external counterevidence and suppressed all four false-positive novelty claims.

2) *Corroboration of the Surviving Hypothesis:* The remaining hypothesis—asserting unresolved **computational burden, memory footprint, and inference latency of ML/DL intrusion detection models on resource-constrained IoT edge devices**—survived all verification checks:

- 1) **Pillar A:** Both OpenAlex and Semantic Scholar queries executed successfully, returning zero confirmed co-mention records bridging lightweight deep learning IDS with hard edge resource constraints under the executed query parameters.
- 2) **Pillar B:** Three independent, source-disjoint 2026 peer-reviewed publications corroborated this explicit limitation:
  - *Publication 8d724497 (2026):* “While Deep Learning based IDS have demonstrated high accuracy, their practical adoption is constrained by high computational overhead, inference latency, and limited interpretability.”
  - *Publication 032c997c (2026):* “While CNN and LSTM networks are widely adopted as benchmarks, their high computational demands pose challenges for real-time deployment on IoT devices.”
  - *Publication 0f849816 (2026):* “Deep learning IDSs have displayed appealing performance; however, they are challenged by heavy computational cost that restricts edge application.”
- 3) **Pillars C & D Integration:** The absence claim is bounded by the empirical extractor recall (77.8%) and contextualized by an unsaturated domain vocabulary ( $\beta = 0.90$ ).

Consequently, rather than falsely declaring an “absolute global gap,” ESV-Gap calibrated its disposition to **Evidence-Supported Gap Hypothesis within an Open Vocabulary Space** (`EVIDENCE_SUPPORTED_BUT_OPEN`).

### D. Autonomous Epistemic Action: Formulated PMCost Agenda

Having validated the candidate, the agent autonomously transitioned from verification to action, formulating a structured, executable PMCost research agenda:

- Problem (P):** Severe computational burdens, memory overhead, and inference latency preventing real-time on-device execution of machine-learning-driven intrusion detection models on edge-layer IoT devices.
- Method (M):** Lightweight convolutional architectures, structured neural network pruning, or self-knowledge distillation for model compression.
- Comparator (C):** Standard uncompressed deep neural network baselines (unpruned CNN, LSTM, and Transformer-based IDS models).
- Outcomes (O):** FLOPs reduction  $\geq 60\%$ , inference latency  $\leq 10$  ms per packet on target hardware, with classification  $F_1$  degradation  $\leq 1.5\%$ .
- Setting (S):** Resource-constrained IoT microcontrollers (e.g., ARM Cortex-M) and edge gateway testbeds.
- Timestamp (T):** September 2026 (Run: `deep_learning_iot_intrusion_de_20260831_114802`).

### E. Status of Comparative Baselines

We explicitly document that comparative empirical baselines—such as benchmarking ESV-Gap against an uncon-

TABLE II  
MULTI-PILLAR EVIDENCE TRIAGE OF POST-GATE CANDIDATE HYPOTHESES

| Candidate Subject<br>( $\mathcal{G}_{\text{subj}}$ ) | Capability<br>Limitation ( $\mathcal{G}_{\text{cap}}$ ) | OpenAlex | S2 Hits  | Pillar A      | Pillar B ( $N_{\text{auth}}$ ) | Final Epistemic<br>Disposition     |
|------------------------------------------------------|---------------------------------------------------------|----------|----------|---------------|--------------------------------|------------------------------------|
| Deep learning IoT IDS                                | Zero-day attack detection                               | 0        | 5        | REFUTED       | 0                              | <b>Refuted</b> (Excluded)          |
| Deep learning IoT IDS                                | Standardized security protocols                         | 0        | 4        | REFUTED       | 0                              | <b>Refuted</b> (Excluded)          |
| Deep learning IoT IDS                                | Industrial testbed validation                           | 0        | 4        | REFUTED       | 0                              | <b>Refuted</b> (Excluded)          |
| ML-driven IoT IDS                                    | Adversarial attack mitigation                           | 0        | 5        | REFUTED       | 0                              | <b>Refuted</b> (Excluded)          |
| <b>ML-driven IoT IDS</b>                             | <b>Computational burden / latency</b>                   | <b>0</b> | <b>0</b> | <b>CORROB</b> | <b>3</b>                       | <b>Evidence-Supported but Open</b> |

strained vanilla LLM (e.g., prompting GPT-4 directly to identify research gaps) or standard dense Retrieval-Augmented Generation (RAG) on this exact gold-standard triage benchmark—were not evaluated in the current physical test run. In Section VIII, we formally specify the experimental protocol and evaluation metrics required to benchmark these alternatives in future work.

## VII. DISCUSSION: IMPLICATIONS FOR AUTONOMOUS MACHINE INTELLIGENCE

### A. From Generative Autonomy to Epistemically Constrained Autonomy

A prevailing assumption in current AI research is that autonomy is synonymous with unconstrained generative power: the more tokens, code files, or hypotheses an agent produces, the more intelligent it appears. Our findings suggest the inverse: **true machine intelligence requires epistemic constraint**. An autonomous agent operating in the real world—whether diagnosing diseases, navigating physical environments, or directing scientific inquiry—must possess the capacity to doubt its own internal representations, challenge its assumptions, and deliberately refuse to act when evidence is lacking.

ESV-Gap demonstrates that epistemic constraint can be operationalized through executable decision contracts. By enforcing fail-closed boundaries, the framework suppressed four spurious hypotheses that a purely generative agent would have confidently promoted. In autonomous systems, the refusal to emit an unverified claim is not a failure of capability; it is the cornerstone of reliability.

### B. Self-Challenging Loops and Active External Grounding

Static representations, whether stored as parametric neural network weights or relational knowledge graphs, are inherently frozen in time and bounded by their ingestion scope. When an autonomous system reasons solely over its internal state, it operates in an epistemic vacuum.

The ESV-Gap Observe–Hypothesize–Verify–Reason–Decide–Act loop establishes a blueprint for self-challenging machine intelligence. Rather than seeking information that confirms its initial hypothesis (confirmation bias), the agent actively

generates adversarial search probes designed to falsify the hypothesis against open-world knowledge indexes. By retrieving recent literature, evaluating token coverage, and examining author disjointness, the agent grounds its reasoning in dynamic external reality.

### C. The Necessity of Graded Epistemic Dispositions

Classical automated reasoning systems typically force binary classifications: a proposition is either true or false; a graph link is either present or absent. Such binarization is brittle in scientific discovery. As shown by our Pillar D analysis, the terminology of deep-learning IoT security is actively expanding ( $\beta = 0.90$ ). Claiming that an absence under such conditions constitutes definitive proof of a global scientific void is epistemically untenable.

By defining graded epistemic states—distinguishing between refuted claims, supported hypotheses within saturated domains, supported hypotheses within open domains, and uncertain candidates requiring abstention—ESV-Gap provides a mathematical and operational framework for calibrated machine assertions. Autonomous systems must communicate not only what they infer, but the precise epistemic boundaries of that inference.

## VIII. THREATS TO VALIDITY

To preserve scientific rigor, we comprehensively document ten threats to the validity of this study:

- 1) **External Index Incompleteness:** OpenAlex and Semantic Scholar index hundreds of millions of scholarly publications, but they do not capture paywalled corporate whitepapers, classified technical reports, non-indexed regional conference proceedings, or preprints hosted exclusively on independent servers. An absence in these indexes does not represent absolute cosmic absence.
- 2) **Query Formulation Sensitivity:** Deterministic boolean queries ( $\mathcal{Q}(h) = u \text{ AND } v$ ) rely on canonical entity keywords. If the broader scientific community addressed the exact problem using completely divergent terminology, the external query probe may fail to retrieve counterevidence.
- 3) **Abstract-Level Semantic Truncation:** External verification was performed over titles and abstracts. While the

primary findings of computer science papers are customarily documented in their abstracts, niche edge deployments or auxiliary efficiency evaluations documented solely in the full-text body or appendix could escape detection.

- 4) **Entity Canonicalization and Synonym Drift:** Although the framework incorporates NFKC normalization, type stripping, and token-based coverage, complex technical multi-word expressions can experience terminology drift over time, potentially impacting multigraph connectivity.
- 5) **Relation Extractor Fidelity:** The empirical recall of the relation extractor was measured at 77.8%, leaving a 22.2% miss rate. Furthermore, the gold benchmark revealed asymmetric performance across relation types, with rare relations such as `CONTRADICTS` exhibiting zero recall on a single sample.
- 6) **Diagnostic Benchmark Scale:** The gold-standard benchmark  $\mathcal{D}_{\text{gold}}$  consists of 36 relation triples across 10 expert-annotated abstracts. While sufficient as an empirical diagnostic to demonstrate that extraction error is non-zero, it represents a limited sample that cannot establish universal extractor bounds across all scientific subdisciplines.
- 7) **Corpus Unsaturation Dynamics:** The fitted Heaps’ law exponent ( $\beta = 0.90$ ) and marginal discovery decay (15.8%) confirm that the 150-paper corpus is not saturated. All positive findings must remain strictly qualified by this open-world reality.
- 8) **Domain Specificity:** The empirical evaluation was conducted within IoT cybersecurity. While the Four-Pillar architecture and mathematical formulation are domain-agnostic, cross-domain performance in biomedical sciences, chemistry, or materials science remains unverified.
- 9) **Temporal Epistemic Validity:** Scientific literature is continuously expanding. A candidate verified as open in September 2026 may be resolved by publications appearing in October 2026. Epistemic decisions are timestamped snapshots, not permanent truths.
- 10) **Absence of Empirical Baseline Comparisons:** This study evaluated the internal validity and triage fidelity of the ESV-Gap pipeline, but did not execute side-by-side comparative trials against vanilla LLM prompting or standard RAG baselines. Future comparative evaluations must measure decision precision, false-positive suppression rate, autonomous abstention rate, and provenance completeness.

## IX. CONCLUSION

We have presented **ESV-Gap**, an auditable autonomous scientific reasoning framework that bridges the critical divide between generative hypothesis ideation and trustworthy evidence verification. Addressing the core challenge of Autonomous Machine Intelligence—enabling artificial agents to reason, verify, and act reliably in open, uncertain environments—ESV-Gap replaces naive link-absence assumptions with an executable epistemic loop: **Observe** → **Hypothesize** → **Verify** → **Reason** → **Decide** → **Act/Abstain**.

Through its Four-Pillar Evidence Architecture, the framework combines active external counterevidence acquisition, source-

disjoint author corroboration, empirical extraction reliability quantification, and corpus saturation diagnostics into a fail-closed decision contract. In an empirical case study on deep-learning IoT intrusion detection, ESV-Gap demonstrated decisive autonomous false-positive suppression, successfully refuting four candidate hypotheses that appeared novel within local corpus boundaries, while provisionally validating an independently corroborated research gap in edge-device computational efficiency. By recognizing its own extraction bounds (22.2% miss rate) and corpus vocabulary expansion ( $\beta = 0.90$ ), the system avoided ungrounded absolute assertions, qualifying the surviving candidate as an evidence-supported hypothesis in an open domain.

Ultimately, ESV-Gap demonstrates that the path toward dependable Autonomous Machine Intelligence lies not in unchecked generation, but in principled epistemic restraint: the capacity of an intelligent system to challenge its own hypotheses, quantify its own uncertainty, and autonomously abstain when evidence is insufficient.

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